Bi-Weekly Status Report

Project Name: Code Sentinel – A Multi-Agent Code Review Platform
Report Period: 03 Sep 2026 – 16 Sep 2026
Prepared By: Aayush Soni (Backend & Database – on behalf of the team)
Date Submitted: 16/09/2026

Sprint/Milestone	Backend Foundation work package complete; Orchestration work package pulled forward and in progress; M5 (core pipeline) due 09 Oct
% Complete (overall)	38%

Indicator	Current	Previous	Trend	Status
Overall Project	38%	25%	↑	On Track
Schedule	95%	100%	↓	Monitor
Scope	100%	100%	→	On Track
Quality	223 tests (CI green)	N/A (pre-code)	↑	On Track
Team Contribution	100%	100%	→	On Track
Dependencies	100%	100%	→	On Track

Summary Notes:
This was the project's first code-producing period and the repository now holds a working monorepo: the Turborepo build, a green CI pipeline on Node 20 and 22, a shared contracts package, the API Gateway and the LangGraph orchestrator – roughly 3,400 lines of source covered by 223 tests across 28 test files. The Backend Foundation work package closed on plan, and the Orchestration work package (planned for 28 Sep–02 Oct) was pulled about two weeks forward: parallel fan-out, per-agent timeouts, de-duplication, severity ranking and status derivation are already implemented and tested. Two items did not go to plan and are the reason the schedule indicator moves down. PostgreSQL migrations and seed scripts (15–16 Sep) were deferred in favour of in-memory stores behind the same interfaces, and the Security and Style agent tasks that were due to start 14 Sep have not begun. Separately, the LLM provider decision was consolidated to Gemini alone, replacing the OpenRouter/Groq/Ollama fallback chain recorded in the earlier deliverables. The team is well placed for M5 on 09 Oct, but agent services are now the critical path.

Key Accomplishments in Two Weeks

- Monorepo and CI complete – npm workspaces with Turborepo, shared ESLint flat config and TypeScript base config; CI runs lint, build, typecheck and test on Node 20 and 22, with a single required `ci` check for branch protection
- Shared contracts package (`@code-sentinel/contracts`) added – zod schemas and inferred types mirroring all four OpenAPI files, plus an examples module for tests; 81 test cases keep the schemas aligned with the specification
- API Gateway working end to end for webhook receipt – HMAC-SHA256 signature verification over raw bytes, a nine-step check order, PR file filtering, review-job construction with an idempotency key, and relay to the orchestrator

- Gateway authentication implemented – HS256 session cookies, checked against the store so revocation takes effect immediately; GET /v1/me and GET `/healthz` live
- LangGraph orchestrator graph running – `fanOut` → `aggregate` → `context` → `threshold`, with parallel agent calls, per-agent timeout, de-duplication, severity ranking and completed/partial/failed status derivation
- 223 test cases across 28 files (contracts 81, gateway 97, orchestrator 45), all running without a network, a real secret or a live agent service
- Large-diff and LLM quota plan written (`plans/large-diffs.md`) and the contracts updated for it – skipped files, review coverage and per-call deadlines

Planned vs. Actual

Task/ Deliverable	Planned Completion	Current Forecast	Actual Status	Notes
Monorepo (Turborepo) + shared configs	Aug 31–Sep 1	—	Done	Single `turbo build` across all workspaces
CI pipeline (GitHub Actions)	Sep 2	—	Done	Green on `main`; Node 20 and 22 matrix; `ci` is the required check
Shared contracts package	Not planned	—	Done	Added this period; removes duplicated types across services
API Gateway scaffold + auth middleware	Sep 3–Sep 6	—	Done	Session cookie scheme; 401 paths tested
Gateway routing + webhook endpoint	Sep 7–Sep 11	—	Done	Signature verification, file filter, orchestrator relay
PostgreSQL schema design	Sep 14	—	Done	Delivered with M4; DDL in `docs/design/schema`
PostgreSQL migrations + seed scripts	Sep 15–Sep 16	Sep 30	Not Started	Deferred; in-memory stores implement the same interfaces
Orchestrator graph, fan-out, timeout	Sep 28–Oct 6	—	Done	Pulled about two weeks forward
Aggregation, de-dup, severity ranking	Sep 28–Oct 2	—	Done	Pulled forward; unit tested
Security Agent: SAST rule integration	Sep 14–Sep 18	Sep 25	Not Started	Slipped; now the critical path to M5
Style Agent: language dispatcher	Sep 14–Sep 17	Sep 24	Not Started	Slipped; same cause

Project Risks, Issues, & Dependency

Type (ID)	Description	Impact	Owner	Mitigation/Next Step
Issue (I-01)	Security and Style agent tasks due to start 14 Sep have not begun; both are MVP scope for M5 on 09 Oct	High	Tej Prakash Tak, Nevil Nandasana	Orchestrator and contracts are already done, so the agent contract and a stub client are available; both tasks start immediately with a shared agent scaffold
Issue (I-02)	PostgreSQL migrations and seed scripts (15–16 Sep) deferred	Medium	Ayush Soni	In-memory stores implement the same interfaces, so swapping in PostgreSQL is contained; rescheduled to 30 Sep
Risk (R-01)	Gemini free-tier quota (15 RPM, 250K TPM, 500 RPD) could throttle reviews once agents call the model	Medium	Sumit Goyal	Quota budgeting designed in `plans/large-diffs.md`: reserve before fan-out, 80% headroom, separate dev and production keys; `src/budget/` to be built next period
Risk (R-03)	Implementation drifting from the OpenAPI contracts	Low	Ayush Soni	Largely closed: `packages/contracts` gives one zod definition per schema, imported by every service and covered by 81 tests
Risk (R-04)	Agent services assume a working gateway and orchestrator	Low	Vatsal Kamani	Closed: both are implemented, tested and callable through the published contract
Risk (R-05)	Superseded deliverables (BRD, Proposal, first report) still describe the multi-provider fallback chain	Low	Sumit Goyal	System Design Document already updated and re-exported; this report records the change for the remaining documents
Dependency (D-01)	GitHub App credentials and webhook secrets needed before Client Integrations (Week 13)	Low	Vatsal Kamani	Unchanged; registration steps documented, to be actioned closer to Week 13

Project Decisions

Decision	Rationale	Impact	Decided By	Date
Gemini is the only LLM provider, for both review calls and embeddings	One provider to budget, test and key; the fallback chain was untestable inside the free tiers	Removes the OpenRouter/Groq/Ollama chain from scope; quota planning becomes mandatory	Team	Sep. 15
Large diffs are skipped, never split	Splitting a file across calls loses the context a review depends on	Oversized files are reported as skipped with a reason instead of silently missed	Sumit, Durgesh	Sep. 15

Decision	Rationale	Impact	Decided By	Date
Shared `packages/contracts` package (zod) added, with the OpenAPI files as source of truth	Every service validating with its own copy of a type is how contracts drift	One definition per schema across all services; 81 tests keep it aligned	Ayush, Sumit	Sep. 08
In-memory stores behind store interfaces until PostgreSQL lands	Unblocks gateway work without waiting on migrations	Migrations move to 30 Sep; no test needs a database	Ayush	Sep. 15
Credentials stored as SHA-256 hashes, not bcrypt	Keys are high-entropy random strings, and a salted hash breaks the unique-hash lookup the schema uses	Documented in the gateway README so the choice is not mistaken for an oversight	Ayush	Sep. 05
Single `ci` job required for branch protection	GitHub counts a skipped required check as passing, so a matrix leg alone is not safe	One required check that fails unless every matrix leg succeeded	Vatsal	Sep. 02

Project Scope Changes

Change	Action	Reason	Schedule Impact	Approved By
LLM provider fallback chain (OpenRouter, Groq, Ollama)	Removed	Consolidated on Gemini; the chain could not be exercised meaningfully within free-tier limits	Frees the 09 Oct fallback-chain task; adds LLM quota budgeting work in its place	Team
Shared contracts package	Added	Prevents contract drift across five services built by different members	Absorbed within the period; no milestone moved	Team
LLM quota budgeting (`src/budget/`, `packages/llm`)	Added	Required once Gemini free-tier limits became the single constraint	Scheduled into the orchestration work package	Team

Upcoming Work (Next Two Week's Plan)

- Start the Security Agent (SAST rule integration, then secret detection) against the published agent contract (Tej Prakash Tak)
- Start the Style Agent language dispatcher and Docker sandbox (Nevil Nandasana)

- Build `src/budget/` in the orchestrator – token estimation, review planning and the quota reservation loop (Sumit Goyal)
- Build `packages/llm` with the Gemini provider and embedder, including the 429 handling path (Sumit Goyal, Durgesh)
- Add the review-job HTTP controller and in-memory job store so the gateway can drive a real review end to end (Sumit Goyal)
- PostgreSQL migrations and seed scripts from `schema.sql`, replacing the in-memory stores (Ayush Soni, by 30 Sep)
- Begin the Next.js dashboard shell against the wireframes and the gateway contract (Moksh Mehta)
- Drive towards M5 on 09 Oct: gateway, orchestrator and two agents connected end to end

Team Member Contributions

Name: Sumit Goyal
Role: Orchestration – LangGraph workflow & LLM abstraction layer

Tasks Completed	LangGraph review graph, parallel fan-out with per-agent timeout, agent HTTP client and agent config loading; large-diff and Gemini quota plan authored
Tasks In Progress	`src/budget/` quota planning and the review-job HTTP controller
Contribution Type	Development / Architecture
Blockers	None
Hours/Capacity	70/200 hrs logged
Evidence	`services/orchestrator/src/graph`, `src/agents`, `plans/large-diffs.md`

Name: Durgesh
Role: Orchestration – Aggregation, severity ranking & Vector DB integration

Tasks Completed	Finding aggregation and de-duplication, severity ranker, review summary counts, status derivation and skipped-file merge, with unit tests
Tasks In Progress	Chroma vector repository implementation and the Gemini embedder
Contribution Type	Development
Blockers	None
Hours/Capacity	68/200 hrs logged
Evidence	`services/orchestrator/src/aggregation` (45 orchestrator test cases), `src/context`

Name: Tej Prakash Tak
Role: Agent Services – Security Agent & Logic Agent

Tasks Completed	Agent contract validated against the orchestrator client; seeded vulnerable-code samples and Semgrep/Bandit rule selection finalized
Tasks In Progress	Security Agent SAST rule integration – start slipped from 14 Sep, now underway
Contribution Type	Research / Development
Blockers	None; agent scaffold now available
Hours/Capacity	56/200 hrs logged
Evidence	`docs/design/openapi/agent.yaml`, rule selection and fixture notes

Name: Nevil Nandasana
Role: Agent Services – Style Agent

Tasks Completed	Language dispatch mapping confirmed against the gateway file filter (`js/ts` → ESLint, `py` → Ruff); sandbox base image selected
Tasks In Progress	Style Agent language dispatcher – start slipped from 14 Sep, now underway
Contribution Type	Research / Development
Blockers	None
Hours/Capacity	55/200 hrs logged
Evidence	`services/gateway/src/webhooks/file-filter.ts` language mapping, sandbox notes

Name: Parin Makwana
Role: Agent Services – Performance Agent & Documentation Agent

Tasks Completed	Review state machine reconciled with the implemented `deriveStatus()` rules; confidence-threshold behaviour confirmed against the threshold node
Tasks In Progress	Hot-path and missing-docstring fixtures ahead of the 19 Oct build start
Contribution Type	Documentation / Research
Blockers	None
Hours/Capacity	52/200 hrs logged
Evidence	`state_review.mmd`, orchestrator status rules, fixture notes

Name: Ayush Soni
Role: Backend & Database – API Gateway & PostgreSQL

Tasks Completed	Gateway app, config validation, structured request logging, error handling, both auth schemes, `GET /v1/me`, `GET /healthz`, webhook receipt with signature verification, file filter, review-job builder and orchestrator client
Tasks In Progress	PostgreSQL migrations and seed scripts – deferred from 16 Sep to 30 Sep
Contribution Type	Development
Blockers	None
Hours/Capacity	72/200 hrs logged
Evidence	`services/gateway/src` (97 gateway test cases), `services/gateway/README.md`

Name: Vatsal Kamani
Role: Integrations & DevOps – Monorepo, GitHub App & Deployment

Tasks Completed	Turborepo monorepo and npm workspaces, shared ESLint and TypeScript configs, CI workflow on Node 20 and 22 with a single required `ci` check, branch-protection procedure documented
Tasks In Progress	Octokit GitHub client to replace the stub, ahead of the 19 Oct GitHub App work package
Contribution Type	Development / Architecture
Blockers	None
Hours/Capacity	64/200 hrs logged
Evidence	`turbo.json`, `.github/workflows/ci.yml`, `eslint.config.mjs`, root `README.md`

Name: Moksh Mehta
Role: Frontend – Next.js dashboard

Tasks Completed	Wireframes mapped to gateway endpoints and to the contracts types the dashboard will consume; component inventory completed
Tasks In Progress	Next.js dashboard shell – scaffolding ahead of the Week 15 build
Contribution Type	Documentation / Development
Blockers	None
Hours/Capacity	58/200 hrs logged
Evidence	`docs/design/wireframes`, `packages/contracts/src/gateway.ts` mapping notes